

BUILT WITH CLAUDE · LIFE SCIENCES · BUILDER TRACK

Onboard a **lab**, not a user.

Point Claude at a lab's messy storage **and** its people. A dynamically-scaled swarm of agents reads everything semantically and folds it into **one shared, cited knowledge base** — that a whole team reads from and writes to, each person driving their own agent, every claim traceable to its source. Including the numbers that live **only inside a figure panel**.

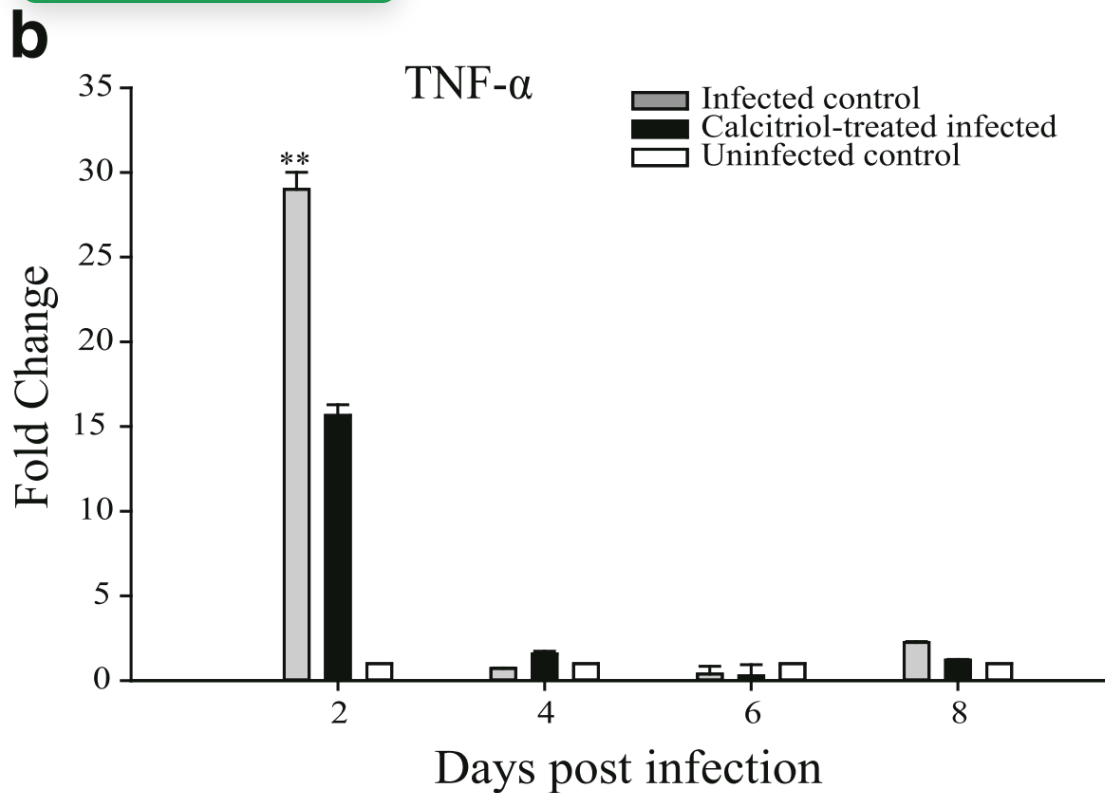
harness · **Claude Science**

+ **Claude Code**

install · **minutes**

license · **MIT**

```
# one line: your team's Claude gets the lab's skills + conventions
$ git clone .../lab-brain && python bootstrap.py --members "Bobby:mentor, Alex:mentee, John:pi"
```



Gui et al. 2017 · Virology Journal · Fig 5b · CC-BY

Claude read this panel blind → TNF- α = 29-fold at day 2.
That number appears nowhere in the paper's text.

THE PROBLEM

Every lab has a mess, and a team — and tools solve for neither.

Years of data, code, decks, docs, grants and papers scatter across Box, Drive and laptops, named 1002732.pdf and 20260210_flu_ABM_BN.pptx. A new member takes months to learn what the lab already knows. And the moment two people — or two AI harnesses — touch the same shared notes, work gets clobbered. "Chat with my files" onboards a *user*. It cannot onboard a *lab*.

Unreadable by design

Cryptic filenames, duplicated dumps, `\_processed/` conventions, two domains interleaved, figures divorced from the code that made them. The knowledge is there — just not findable.

02 · the team

Contribution collides

Peers and undergrads each know a slice. Without a shared convention, concurrent edits overwrite each other and provenance evaporates. Onboarding a whole team is a coordination problem, not a chat.

THE APPROACH

Three moves that turn storage + people into one brain.

move 01

Onboard fast

One command wires a newcomer's Claude Code *and* Claude Science to the shared KB, with the lab's conventions preloaded as skills. Zero to connected in minutes.

move 02

Contribute, don't just read

A swarm of area-specialist agents reads each artifact in full, verifies, and writes to the KB with provenance and a confidence — never a skim.

move 03

A whole team, one brain

Peers and mentees contribute concurrently through a shared **framework of conventions** — where things go, how figures are made, how mentees report up. Additive, no clobbering. **That framework is the hero.**

THE CORE · YOUR MESS → YOUR BRAIN

Point it at a lab's files. It reads everything, and files each where it belongs.

Code, slides, docs, grants, data, papers — however cryptic or disorganized. A cheap **structural sort** (runnable today, no LLM) buckets every file, catches duplicates, and drops non-science; then a **swarm** of area-specialist agents reads each file in full and routes what it learns into the shared KB. **Click a file from a real messy lab folder.**

13 files → **7** content areas · **1** duplicate caught · **admin folder** excluded ·

temp files skipped

a real run of `python -m labbrain.lab_scan`

code/20260210_flu_ABM_BN.py

code

20251120_kickoff_JWH.pptx

slides

R01_specific_aims_v3.docx

grant

raw_counts.csv

data

1002732.pdf

paper · cryptic name

_processed/raw_counts.csv

duplicate

Admin/budget_2026Q1.xlsx

not this project

READS

imports, the files it reads/writes, entrypoint, tests (via the code AST)

EXTRACTS

reads raw_counts.csv → writes results/fit.json; entrypoint run()

→ GOES TO

a code record in the registry + a code→data edge in the provenance graph

THE FRAMEWORK • THE HERO

"Where does this go?" — answered by default.

A lab loses knowledge not because people don't write things down, but because nobody agrees on *where things go* — so updates, decisions, and findings scatter into chat logs and vanish. The framework gives every artifact one home. **Pick something you just did.**

Finished something / hit a blocker today

Need a teammate (or the other harness) to pick something up

Did an ingest / a significant write

Made a figure

Wrote a how-to / an SOP / synthesis

Pulled in a paper / transcript / data dump

A mentee finished a round of work

Learned a durable fact worth recalling

→ GOES TO daily brief

START_HERE.md

The rolling daily inbox everyone opens first — dated, with a     status. Links + one-liners only; the detail lives in the actual file.

SIX CONVENTIONS (+ TWO)

Generalized from a system a real lab already runs.

01 · files

KB structure

Three layers — **Sources** (raw, immutable) · **Wiki** (agent-owned synthesis) · **Output** (private strategy) — and one filing rule so evidence and interpretation never blur.

02 · names

Naming & notes

A trailing-YYYY-MM-DD convention so filename-sort = sense, plus one YAML note template every file starts from.

03 · docs

Documentation & handoffs

The routing map above: START_HERE daily-brief (   ) , _Log, the _handoff-log board, MEMORY.

04 · figures

Figures & findings

Restyle, never reinvent; verify by looking. Scripts read real data (no hard-coded numbers), and FIGURE_FINDINGS.md logs what each panel shows. Captions are **claims, not labels**.

05 · storyline

Storyline & manuscript

Messages first, figures follow. A per-figure storyline doc where every panel earns a “📖 **Beat**” in the argument, kept in sync with the prose.

06 · onboarding

Onboarding a member

A walkable sequence + a per-person scaffold with **supervision gates** and an **Unknowns Register** that extracts the mentor's tacit knowledge instead of losing it.

+ **mentorship** – the mentee↔mentor check-in loop (next section) · + **harness playbook** – Claude Science & Claude Code on one KB: one harness owns a working-dir at a time.

see it lived-in

The framework, in use — not blank templates

The repo ships `examples/demo-lab-vault/`: a realistic populated vault (a lab onboarding a mentee onto a flu-cytokine sub-track, grounded in the real Gui 2017 paper) — a daily-brief `START_HERE`, a coordination log, a supervision-gated onboarding scaffold + a filled Unknowns Register, a paired check-in round, and a `FIGURE_FINDINGS` log.

DEPTH · SCALES TO THE WORK

The number of agents adapts to the corpus.

Nobody hand-tunes it. A 200-file lab gets a handful of readers; a 12,000-file lab gets tens, fanned out concurrently — one specialist per content area, bounded so it never runs away. Drag the corpus size.

corpus size  1,400 files

spawned **67** readers across 8 content areas

code 12

slides 10

lit 12

docs 10

data

8

comms

6

grants

4

media

5

```
W_area = clamp( ceil(files_in_area / files_per_worker), 1, max_workers ) · bounded by triage  
tiers + a per-run cap
```

MULTI-USER · ALREADY IN PRODUCTION

Peers work as equals. Mentees get scaffolded — and report up.

The roster assigns each member a role, and the collaboration shape follows. The real engine is a **paired-by-date check-in loop**: a neutral – update (what shipped) kept separate from a structured –feedback (the mentor's voice). Flip the mode.

Peers (maintainers)

Mentor ↔ mentee

- ▶ **Symmetric** — everyone reads the shared KB, works a lane, writes back **additively**
- ▶ **Per-user paths** — **resolve(key)** picks the right storage for whoever runs it — the #1 multi-user fix
- ▶ **Reviewed as an equal** — a maintainer's commits are peer contributions — no scaffolding, no digest
- ▶ **No collisions** — claim a lane on **LANES.md**, dated **Sources/** notes, one branch per lane

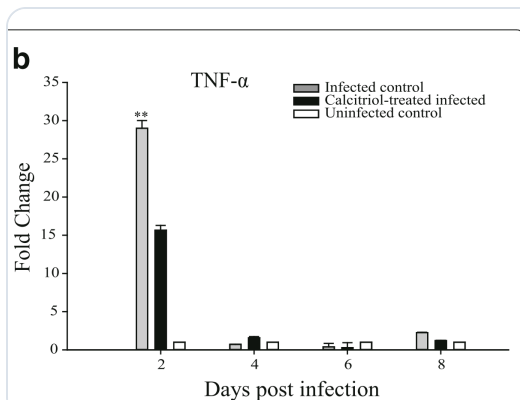
PERSON · LANE	HARNESS	STATUS
@BN · figure slice	CC	active
@AR · flu-index	CS	active
@JWH · gaps audit	—	watching

LANES.md — claim a lane before you work; collisions prevented by convention, not locks.

ONE SHOWCASED CAPABILITY

It even reads the figures — and stays honest.

Reading a paper doesn't stop at the text. Lab Brain crops each figure panel and pulls the numbers out — then flags anything it can't stand behind.



✓ VERIFIED confidence **high**

TNF- α = 29-fold at day 2

a single cropped bar-chart panel · Gui 2017, Fig 5b · CC-BY

This number is stated nowhere in the paper's text — it exists only in the figure. Reading the text doesn't give it to you; reading the panel does.

ALSO CAUGHT, HONESTLY

IL-6 reads ~500-fold but the paper's text says "**5.6 fold**" → the clash is flagged **needs-review**, not silently trusted. A figure with a broken axis is declined rather than guessed. The panel crop is kept as provenance for every value.

provenance · DOI 10.1186/s12985-017-0683-y · CC-BY-4.0 · crop retained

THE COMMAND SUITE · 16 SKILLS

Cheap-structural first, expensive-semantic second — human-gated.

The onboarding pipeline is staged by cost so an expensive crawl never runs before a human approves it — plus framework skills (`/lab-figure` · `/lab-findings` · `/lab-storyline` · `/lab-feedback`) that apply the conventions. Click a pipeline step.

`/lab-init`

`/lab-scan`

`/lab-triage`

`/lab-index`

`/lab-link`

`/lab-ask`

`/lab-standup`

`/lab-update`

`/lab-init` · `configure`

Configure

Interview the lab (or sample a subtree) → one `lab-profile.yaml`: storage, naming, ROSTER+roles, vocabulary, privacy. The whole customization layer.

ALREADY IN THE LAB

"We don't just propose this
— we're onboarding our
own team onto it."

The collaboration model is generalized from the coordination practices the **Hickey Lab runs day-to-day** on a multi-person spatial-omics project — CODEX spatial proteomics × MALDI-IMS lipidomics on human testis — plus a Data+ undergraduate team. The practice is real; Lab Brain packages it so any lab can adopt it — and we're adopting this version ourselves.

2

harnesses,
one brain

8

onboarding
commands

4

roles:
pi · mentor · maintainer · mentee

Peers (maintainers) work symmetric lanes; mentees get extra scaffolding + mentor feedback — from the same shared vault, driven by `/lab-standup`.

INSTALL

Clone, bootstrap, contribute.

```
# 1 · install the kit – skills into your .claude, collab structure into your KB
$ git clone https://github.com/<you>/lab-brain && cd lab-brain
$ python bootstrap.py --members "Bobby:mentor, Alex:mentee, John:pi"

# 2 · in Claude Code – configure, then build the shared brain
> /lab-init → /lab-scan → /lab-index → /lab-link
> /lab-ask "what datasets does the flu project use, and who owns them?"
> /lab-read-figure # the verified figure values
> /lab-standup # per-person digests + a team rollup
```



Lab Brain

MIT

Claude Code + Claude Science

open access data only

